

**SECTION 1. IDENTIFICATION OF THE CHEMICAL AND COMPANY**

Product name : ECO-STAR SOUR PLUS  
 Other means of identification : Not applicable  
 Recommended use : Laundry product  
 Restrictions on use : Reserved for industrial and professional use.  
 Product dilution information : Product is sold ready to use.  
 Company : Chile ECOLAB S.A.  
 Avenida Pedro de Valdivia 3801  
 Santiago, Santiago Chile  
 (2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600  
 sac.chile@ecolab.com  
 Emergency telephone : (+56-2) 2247-3600 (CITUC)  
 Toxicological Emergency Phone : CITUC (+56-2) 2635-3800 (24 hours)

**SECTION 2. HAZARDS IDENTIFICATION**

**Classification according to NCH 382** : Class: Class 8 Corrosives

**Distintivo según NCh 2190** :


**GHS Classification**

Acute toxicity (Oral) : Category 5  
 Skin corrosion : Category 1  
 Serious eye damage : Category 1

**GHS label elements**

**Hazard pictograms** :



**Signal Word** : Danger

**Hazard Statements** : May be harmful if swallowed.  
 Causes severe skin burns and eye damage.

**Precautionary Statements** : **Prevention:**  
 Wash skin thoroughly after handling. Wear protective gloves/  
 protective clothing/ eye protection/ face protection.  
**Response:**

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor. Call a POISON CENTER/doctor if you feel unwell. Wash contaminated clothing before reuse.

**Storage:**

Store locked up.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

**Other hazards** : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

### SECTION 3. COMPOSITION/INFORMATION OF INGREDIENTS

Pure substance/mixture : Mixture

| Chemical name      | CAS-No.    | Concentration (%) |
|--------------------|------------|-------------------|
| fluorosilicic acid | 16961-83-4 | 10 - 30           |
| citric acid        | 77-92-9    | 5 - 10            |

### SECTION 4. FIRST AID MEASURES

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

**Expected acute effects**

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns.

Ingestion : May be harmful if swallowed. Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

**Expected chronic effects**

Chronic Exposure : Health injuries are not known or expected under normal use.

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

#### Most important symptoms / effects

|              |                                 |
|--------------|---------------------------------|
| Eye contact  | : Redness, Pain, Corrosion      |
| Skin contact | : Redness, Pain, Corrosion      |
| Ingestion    | : Corrosion, Abdominal pain     |
| Inhalation   | : Respiratory irritation, Cough |

#### SECTION 5. FIREFIGHTING MEASURES

|                                                           |                                                                                                                                                                                 |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extinguishing Agents                                      | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                                                       |
| Unsuitable extinguishing agents                           | : None known.                                                                                                                                                                   |
| Specific hazards during fire fighting                     | : Not flammable or combustible.                                                                                                                                                 |
| Products generated in combustion and thermal degradation  | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus                                |
| Precautions for Emergency Personnel and / or Firefighters | : Use personal protective equipment.                                                                                                                                            |
| Specific extinguishing methods                            | : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes. |

#### SECTION 6. MEASURES IN CASE OF ACCIDENTAL SPILLAGE

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions, protective equipment and emergency procedures | : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.                                                      |
| Environmental precautions                                           | : Do not allow contact with soil, surface or ground water.                                                                                                                                                                                                                                                                                                                                                    |
| Methods and materials for containment and cleaning up               | : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. |

#### SECTION 7. HANDLING AND STORAGE

##### Handling

|                         |                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Advice on safe handling | : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

#### Storage

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Recommended and unsuitable packaging by supplier : Use high density plastic containers.

Storage temperature : 10 °C to 30 °C

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:  
Impervious gloves, resistant to chemicals.  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Wear closed shoes.

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid

Color : clear, violet

Odor : slight

pH : 1,0 - 2,0, (100 %)

Flash point : Not applicable, Does not sustain combustion.

Odor Threshold : No data available

Melting point/freezing point : No data available

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

|                                         |                     |
|-----------------------------------------|---------------------|
| Initial boiling point and boiling range | : > 100 °C          |
| Evaporation rate                        | : No data available |
| Flammability (solid, gas)               | : No data available |
| Upper explosion limit                   | : No data available |
| Lower explosion limit                   | : No data available |
| Vapor pressure                          | : No data available |
| Relative vapor density                  | : No data available |
| Relative density                        | : 1,19 - 1,25       |
| Water solubility                        | : soluble           |
| Solubility in other solvents            | : No data available |
| Partition coefficient: n-octanol/water  | : No data available |
| Autoignition temperature                | : No data available |
| Thermal decomposition                   | : No data available |
| Viscosity, kinematic                    | : No data available |
| Explosive properties                    | : No data available |
| Oxidizing properties                    | : No data available |
| Molecular weight                        | : No data available |
| VOC                                     | : No data available |

### SECTION 10. STABILITY AND REACTIVITY

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical stability                 | : Stable under normal conditions.                                                                                                                |
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas.                                                                |
| Conditions to avoid                | : None known.                                                                                                                                    |
| Incompatible materials             | : Bases<br>Metals                                                                                                                                |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus |

### SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

#### Toxicity

#### Product

|                           |                                         |
|---------------------------|-----------------------------------------|
| Acute oral toxicity       | : Acute toxicity estimate : 2.024 mg/kg |
| Acute inhalation toxicity | : No data available                     |

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

|                                   |                     |
|-----------------------------------|---------------------|
| Acute dermal toxicity             | : No data available |
| Skin corrosion/irritation         | : No data available |
| Serious eye damage/eye irritation | : No data available |
| Respiratory or skin sensitization | : No data available |
| Carcinogenicity                   | : No data available |
| Reproductive effects              | : No data available |
| Germ cell mutagenicity            | : No data available |
| Teratogenicity                    | : No data available |
| STOT-single exposure              | : No data available |
| STOT-repeated exposure            | : No data available |
| Aspiration toxicity               | : No data available |

### SECTION 12. ECOLOGICAL INFORMATION

#### Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

#### Product

|                                                     |                     |
|-----------------------------------------------------|---------------------|
| Toxicity to fish                                    | : No data available |
| Toxicity to daphnia and other aquatic invertebrates | : No data available |
| Toxicity to algae                                   | : No data available |

#### Ingredients

|                  |                                                 |
|------------------|-------------------------------------------------|
| Toxicity to fish | : fluorosilicic acid<br>96 h LC50 Fish: 50 mg/l |
|                  | citric acid<br>96 h LC50 Fish: > 100 mg/l       |

#### Persistence and degradability

Readily biodegradable.

#### Bioaccumulative potential

No data available

#### Mobility in soil

No data available

#### Other adverse effects

No data available

### SECTION 13. INFORMATION OF DISPOSAL

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

|                                    |                                                                                                                                                                                                                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disposal considerations            | : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.                                                |
| Disposal of contaminated packaging | : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat such waste, which must be issued and final disposal certificate waste. |

#### SECTION 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

##### Land transport

###### Official transport classification of the United Nations

|                           |                      |
|---------------------------|----------------------|
| UN number                 | : 1778               |
| Description of the goods  | : FLUOROSILICIC ACID |
| Class                     | : 8                  |
| Packing group             | : II                 |
| Environmentally hazardous | : no                 |

##### Air transport (IATA)

###### Official transport classification of the United Nations

Contact Regulatory for air freight eligibility

##### Sea transport (IMDG/IMO)

###### Official transport classification of the United Nations

|                                                                          |                      |
|--------------------------------------------------------------------------|----------------------|
| UN number                                                                | : 1778               |
| Description of the goods                                                 | : FLUOROSILICIC ACID |
| Class                                                                    | : 8                  |
| Packing group                                                            | : II                 |
| Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code | : Not applicable     |
| Marine pollutant                                                         | : no                 |

#### SECTION 15. REGULATORY INFORMATION

##### Registrations and Certifications

Chile: Our HDS meets Chilean Normative: NCH 2245: 2015 (Safety Data Sheet for chemical products - Content and order of sections)

The receiver should pay attention to the possible existence of local regulations.

NCh 1411: Prevention of risks, IV identification of risks of materials

NCh 2190: Transportation of Hazardous Substances - Pictograms for hazard identification

NCh 382: Dangerous Good - Classification

D.S. No. 594: Minimum basic conditions in the workplace

D.S. No. 148: Hazardous waste disposal

D.S. No. 72: Mining Safety Regulations

D. S. No. 43: reports on the storage of hazardous substances

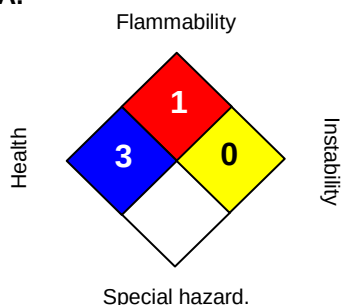
D. S. No. 40: Reporting on exposure hazards

#### SECTION 16. OTHER INFORMATION

## SAFETY DATA SHEET

### ECO-STAR SOUR PLUS

#### NFPA:



#### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 3 |
| FLAMMABILITY    | 1 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

Issuing date : 06.12.2017  
Version : 1.0  
Prepared by : Regulatory Affairs

While there is no change in the formula or hazard classifications, this SDS remains valid

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.